

Signatures of Shock Drivers in the Solar Wind and Their Dependence on the Solar Source Location

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Solar wind and energetic ion observations following 40 interplanetary shocks with well-established solar source locations have been examined in order to determine whether signatures characteristic of the coronal material forming the shock driver are present. The signatures considered include magnetic-field-aligned bidirectional ion flows observed by the ISEE 3 and IMP 8 spacecraft; bidirectional solar wind electron heat fluxes; solar wind plasma proton and electron temperature depressions; low- β plasma; enhanced, low-variance magnetic fields; and energetic ion depressions. Several shock driver signatures are commonly observed following shocks originating from within $\sim 50^\circ$ of central meridian and are generally absent for other events. We conclude that shock drivers generally extend up to $\sim 100^\circ$ in longitude, centered on the solar source longitude. A particular shock driver signature is typically found following $\sim 80\%$ of shocks from near-central meridian events. However, only $\sim 20\%$ of shocks from near-central meridian events are associated with the bidirectional 35–1000 keV ion events discussed by Marsden *et al.* (1987). Since shocks from central meridian events are not usually associated with all the shock driver signatures examined, the absence of a driver cannot be confirmed from consideration of one of these signatures alone. We also find evidence that a few bidirectional energetic ion and solar wind electron heat flux events following shocks (in particular from far eastern sources) may occur on open field lines outside of shock drivers.

1. INTRODUCTION

It is well known that a coronal mass ejection (CME) may drive a shock if it moves rapidly enough through the solar wind [Sheeley *et al.*, 1983, 1985]. Various observations in the interplanetary medium have been interpreted as the interplanetary signatures of CMEs [Gosling, 1990, and references therein]. These signatures include depressed temperatures of solar wind plasma electrons [Montgomery *et al.*, 1974] and protons [Gosling *et al.*, 1973], enhanced solar wind helium abundances [Hirshberg *et al.*, 1972; Borrini *et al.*, 1982], magnetic-field-aligned bidirectional solar wind electron heat fluxes [Gosling *et al.*, 1987, 1990] and bidirectional energetic ion flows [Rao *et al.*, 1967; Palmer *et al.*, 1978; Kutchko *et al.*, 1982; Sarris and Krimigis, 1982; Sanderson *et al.*, 1983; Marsden *et al.*, 1987; Tranquille *et al.*, 1987; Richardson and Reames, 1991, 1993], and low-variance magnetic field enhancements [Burlaga *et al.*, 1981; Gosling *et al.*, 1987; Marsden *et al.*, 1987; Tranquille *et al.*, 1987] that often include smooth rotations of the magnetic field direction in so-called “magnetic clouds” [Klein and Burlaga, 1982; Lepping *et al.*, 1990]. The plasma β (ratio of the particle to magnetic field pressures) is often unusually low [Burlaga *et al.*, 1981; Gosling *et al.*, 1987]. Solar wind ions with anomalous ionization states may also be present [Zwickl *et al.*, 1982]. Where these phenomena follow interplanetary shocks, they are interpreted as evidence of entry into CME material forming the shock driver. Depressions in the energetic ion intensity following shocks have also been associ-

ated with entry into the shock driver [Marsden *et al.*, 1987; Cane *et al.*, 1988]. It has been proposed that shock drivers include closed magnetic field topologies [Piddington, 1958] or magnetic field loops rooted at the Sun [Morrison, 1954; Cocconi *et al.*, 1958; Gold, 1959] and that the bidirectional particle flows originate from particle trapping and reflection within these regions [Palmer *et al.*, 1978; Kutchko *et al.*, 1982; Sarris and Krimigis, 1982]. The depressed plasma temperatures may result from cooling of these expanding regions as they move away from the Sun [Gosling *et al.*, 1973; Montgomery *et al.*, 1974].

Since not all shocks are followed by driver signatures, it has been concluded that shocks extend over larger longitudinal ranges than their drivers [Borrini *et al.*, 1982; Cane, 1988]. However, as noted by Zwickl *et al.* [1983], a particular shock driver signature may be absent following a shock even though other signatures are clearly evident. Therefore, ideally, several signatures should be considered together in order to assess more accurately whether a driver is absent.

In this paper we examine evidence for the presence of several types of driver signatures following 40 energetic shocks in 1978 to 1982 with well-established solar source regions [Cane, 1985]. These signatures include recently surveyed observations of bidirectional ~ 1 MeV/amu ion flows detected by the Goddard Space Flight Center (GSFC) instruments on ISEE 3 and IMP 8 [Richardson and Reames, 1991, 1993]. The data used in this study are outlined in section 2. Section 3 discusses the association between the shock solar source longitude and the presence or absence of shock driver signatures, while three interesting events are described in section 4. The observations are discussed in section 5.

2. SHOCK DRIVER SIGNATURE DATA SOURCES

The shock driver signature observations used in this paper were obtained largely from the IMP 8 and ISEE 3 spacecraft.

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IMP 8, in a 12.5-day period geocentric orbit, was in the solar wind upstream of the Earth's bow shock for $\sim 60\%$ of each orbit. During all but one of the events discussed by Cane [1985] and listed in Table 1, ISEE 3 was in orbit around the L1 libration point, 1.5×10^6 km upstream of Earth, where it monitored the near-Earth solar wind continuously with high data coverage ($>90\%$). The exception, the shock on February 4, 1983, occurred during the ISEE 3 geotail mission [Tsurutani and von Rosenvinge, 1984] when the spacecraft was $\sim 220 R_E$ downtail of Earth. ISEE 3 exited from the geomagnetic tail following shock passage and detected high-speed (~ 900 km/s) solar wind plasma in the magnetosheath [Fairfield and Phillips, 1990].

Magnetic-field-aligned bidirectional ~ 1 MeV/amu ion events have been identified using data from the GSFC instruments on IMP 8 and ISEE 3 (as discussed by Richardson and Reames [1993]). Reports of bidirectional 35–1000 keV ion events from the Energetic Particle Anisotropy Spectrometer instrument on ISEE 3 [Marsden et al., 1987] and of bidirectional solar wind electron heat flux events detected by the Los Alamos National Laboratory solar wind electron plasma experiment on ISEE 3 [Gosling et al., 1987, 1990; J. T. Gosling, private communication, 1989] have also been considered. The presence of postshock ion intensity depressions is inferred using 1–20 MeV ion data from the IMP 8 and ISEE 3 spacecraft. Near-Earth solar wind data from the National Space Science Data Center (NSSDC) OMNI data base [Couzens and King, 1986] have been examined to infer the presence of plasma proton temperature depressions, while data from the ISEE 3 magnetometer have been used to identify low-variance magnetic field enhancements. Solar wind electron data have been obtained from hourly averaged observations from the ISEE 3 electron plasma experiment provided by NSSDC.

We do not consider the presence of helium abundance enhancements or anomalous ionization states because there are no complete studies identifying events during our interval of study. Cosmic ray Forbush decreases are also characteristic of many shock drivers [Barnden, 1973; Sanderson et al., 1990]. However, since Forbush decreases may arise from turbulent postshock regions as well as from shock drivers [Cane, 1993], they also will not be considered in this study. Likewise, we will not consider the presence of magnetic field rotations which, while they are characteristic of a subset of shock drivers (in particular, magnetic clouds), are clearly absent from many regions exhibiting other signatures of CME-associated material [Gosling, 1990].

3. DETECTION OF SHOCK DRIVER SIGNATURES VERSUS SOURCE LONGITUDE

As discussed by Cane [1985], the solar source locations for the shocks in Table 1 (given in terms of the longitude from central meridian and latitude) have been inferred from interplanetary type II radio emission generated by the shocks as they pass through the interplanetary medium from the source. The shock time shown is the time of the geomagnetic storm sudden commencement (SC) associated with shock passage at Earth. Typically, the shocks passed ISEE 3 at the upstream libration point ~ 30 min earlier.

Table 1 also indicates whether various driver signatures were evident commencing within ~ 24 hours of shock passage (corresponding to within ~ 0.25 AU downstream of the

shock for typical solar wind speeds). Columns 3 and 4 indicate whether or not ISEE 3 and IMP 8 bidirectional ~ 1 MeV/amu ion flows (BIFs) with minimum durations of 30 min were observed, while the next two columns show the presence of bidirectional solar wind electron heat fluxes (BHFf) and 35–1000 keV ion bidirectional flows. Three center dots ($\cdot \cdot \cdot$) in Table 1 indicate that the data are not available or contain significant gaps or, in the case of the IMP 8 bidirectional ion flows, that the spacecraft was not in the solar wind. Column 7 indicates whether a region was present in which the plasma proton temperature T_p was anomalously depressed, in particular, by at least a factor of 4 below that expected from the well-established general relationship between solar wind speed and proton temperature [Lopez, 1987, and references therein]. Note that for many events in 1980–1982 the OMNI proton temperature data are not available or are incomplete. An electron temperature T_e depression is indicated in column 8 if the temperature fell below 10^5 K. (This temperature is exceeded by $\sim 80\%$ of ISEE 3 plasma electron temperature measurements made in 1978–1982.) Column 9 indicates whether plasma with a low value of β ($= 2\mu_0(n_e kT_e + n_p kT_p)/B^2$) was present. We define “low β ” here as $\beta \leq 0.2$, a level which is exceeded by $\sim 95\%$ of the plasma measurements in 1978–1982. (When T_p measurements are incomplete but T_e data are available, we estimate β from the electron pressure, which usually makes the major contribution to the particle pressure since generally $T_e > T_p$.) Column 10 indicates whether a region of strong, quiet magnetic field was present. This is inferred from 15-min averages of the ISEE 3 magnetic field intensity and defined by $B \geq 10$ nT and $\sigma B/B \leq 10^{-2}$, criteria satisfied by only $\sim 4\%$ of the field data. The final signature included in Table 1 is the presence of an abrupt energetic ion intensity depression. In a few cases, unrelated energetic particle onsets make it difficult to determine whether an ion depression was present.

Table 1 clearly supports the finding of Zwickl et al. [1983] that in general, following interplanetary shocks, some driver signatures are present while others are absent. The number of signatures (with data available) observed following each shock is indicated in Table 1, together with the percentage of these signatures observed.

Figure 1 summarizes the relationship between the detection of each driver signature and the solar source longitude. The triangles, plotted versus source longitude, show whether or not each signature was observed (symbol plotted at top or bottom of panel, respectively) where the appropriate data are available. The percentage probabilities of the driver signature being observed following events binned in 30° width intervals of source longitude are also shown.

Although there are variations from signature to signature, and in some cases events are scarce in particular longitude ranges, it is evident from Figure 1 that in general, driver signatures are observed following the majority of shocks with sources around (in particular, within $\sim 50^\circ$ of) central meridian but are absent following shocks originating farther from central meridian. The observations suggest that shock drivers usually extend in heliolongitude up to $\sim 50^\circ$ from the solar source. There are indications of asymmetries in some signatures, but in general, these arise from the limited numbers of events in some longitude ranges. However, some may be significant, as discussed below.

The observability of the driver signatures also varies.

TABLE 1. Driver Signatures Following Cane [1985] Interplanetary Shocks

Year	SC Time	Source Location	ISEE 3 ~1 MeV BIF	IMP 8 ~1 MeV BIF	ISEE 3 BHF	ISEE 3 35-1000 keV BIF	T_p Depressed	T_e Depressed	$\beta < 0.2$	>10 nT Quiet Field	Ion Intensity Depression	Number of Signatures Observed ^a	Percent of Signatures With Data Observed ^b
1978	Sept. 25, 0718	W50, N35	N	N	N	N	N	N	N	N	N	0/9	0
	Oct. 4, 0047	E57, S13	Y	Y	N	N	N	N	N	Y	N	3/9	33
	Nov. 12, 0100	E01, N17	Y	Y	Y	Y	Y	Y	Y	Y	Y	9/9	100
	Dec. 14, 0127	W50, S16	N	...	N	N	Y	Y	N	Y	Y	4/8	50
1979	Feb. 18, 0304	E59, N16	Y	...	N	N	N	N	N	Y	N	2/8	25
	March 4, 0445	E58, S23	Y	...	N	N	N	N	N	Y	N	1/9	11
	March 15, 0556	W76, S24	Y	...	N	N	N	N	N	Y	N	2/8	25
	April 5, 0155	W14, S25	Y	Y	Y	Y	Y	Y	Y	Y	Y	9/9	100
1980	April 24, 2357	E10, S20	Y	Y	Y	N	Y	Y	Y	Y	Y	8/9	89
	July 6, 1930	E36, N11	Y	...	N	N	Y	Y	N	Y	N	4/8	50
	Aug. 20, 0625	E90, N08	N	Y	Y	N	N	N	N	Y	N	2/9	22
	Aug. 29, 0459	W11, N05	Y	Y	Y	N	Y	Y	N	Y	Y	7/9	78
1980	April 6, 1059	W35, N27	Y	...	Y	N	...	...	...	N	N	4/7	57
	July 18, 1926	E06, S11	N	...	Y	N	...	...	...	N	N	1/8	13
	July 25, 1111	E17, S19	Y	...	Y	Y	...	...	...	Y	Y	4/4	100
	Oct. 18, 0114	E55, N21	N	...	Y	N	N	N	N	Y	N	2/9	22
1981	Nov. 14, 1153	W69, S11	N	N	Y	N	N	Y	Y	Y	N	2/9	22
	Nov. 24, 2257	E39, S18	N	N	Y	N	Y	Y	Y	Y	?	4/7	57
	Nov. 26, 0422	W20, N13	N	Y	N	N	Y	Y	N	?	...	3/7	43
	April 3, 0347	W52, S43	N	N	N	N	N	N	N	N	?	0/8	0
1981	April 26, 0813	W50, N14	Y	Y	N	N	Y	Y	N	?	?	4/7	57
	May 10, 2208	E42, N09	N	...	Y	N	...	N	N	Y	N	2/7	29
	May 14, 1856	E54, N13	Y	...	Y	N	...	N	Y	Y	?	4/6	67
	May 16, 0532	E32, N20	Y	...	N	N	...	N	Y	Y	?	3/6	50
1982	May 17, 2302	E14, N10	Y	Y	Y	N	Y	Y	Y	Y	Y	8/9	89
	Aug. 10, 0434	E24, S10	N	...	Y	N	...	N	Y	Y	N	2/7	29
	Aug. 23, 1257	E02, S17	Y	...	N	N	...	Y	Y	Y	N	4/7	57
	Oct. 10, 1434	E83, S13	N	...	Y	N	...	N	N	Y	N	2/7	29
1982	Oct. 13, 2240	E30, S17	Y	...	Y	N	...	N	N	Y	N	1/8	14
	Nov. 11, 1238	E17, S16	N	Y	Y	N	Y	...	Y	Y	Y	7/8	88
	Nov. 25, 0229	W21, N12	Y	Y	Y	N	Y	Y	Y	N	Y	7/9	78
	Dec. 12, 0144	W16, N12	N	N	Y	N	N	Y	N	N	Y	3/9	33
1983	Dec. 29, 0455	E16, S13	Y	Y	Y	N	Y	Y	Y	Y	Y	8/9	89
	Feb. 1, 1100	E13, S16	N	...	Y	N	...	N	Y	Y	?	2/6	33
	Feb. 3, 0129	W09, S14	Y	...	Y	N	...	N	Y	...	...	3/5	60
	July 11, 0953	E76, N17	N	N	Y	...	N	N	N	Y	N	2/8	25
1983	July 13, 1617	E37, N11	Y	Y	Y	...	...	Y	Y	Y	Y	100	100
	Sept. 5, 2250	E33, N13	Y	...	Y	...	...	Y	Y	Y	Y	6/6	100
	Sept. 21, 0339	E06, S16	Y	Y	Y	...	...	Y	Y	Y	Y	7/7	100
	Feb. 4, 1615	W08, S19	Y (MS)	...	...	...	...	...	...	...	Y	2/2	100

SC time is that of the geomagnetic storm sudden commencement associated with shock passage at Earth; source location is given in terms of the longitude from central meridian and latitude; BIF, bidirectional ion flows; BHF, bidirectional heat fluxes; T_p , plasma proton temperature; T_e , plasma electron temperature; N, no; Y, yes; MS, observed in magnetosheath.

^aNumber of signatures observed following each shock and number of signatures with data available.

^bHe indicates helium abundance enhancement following shock.

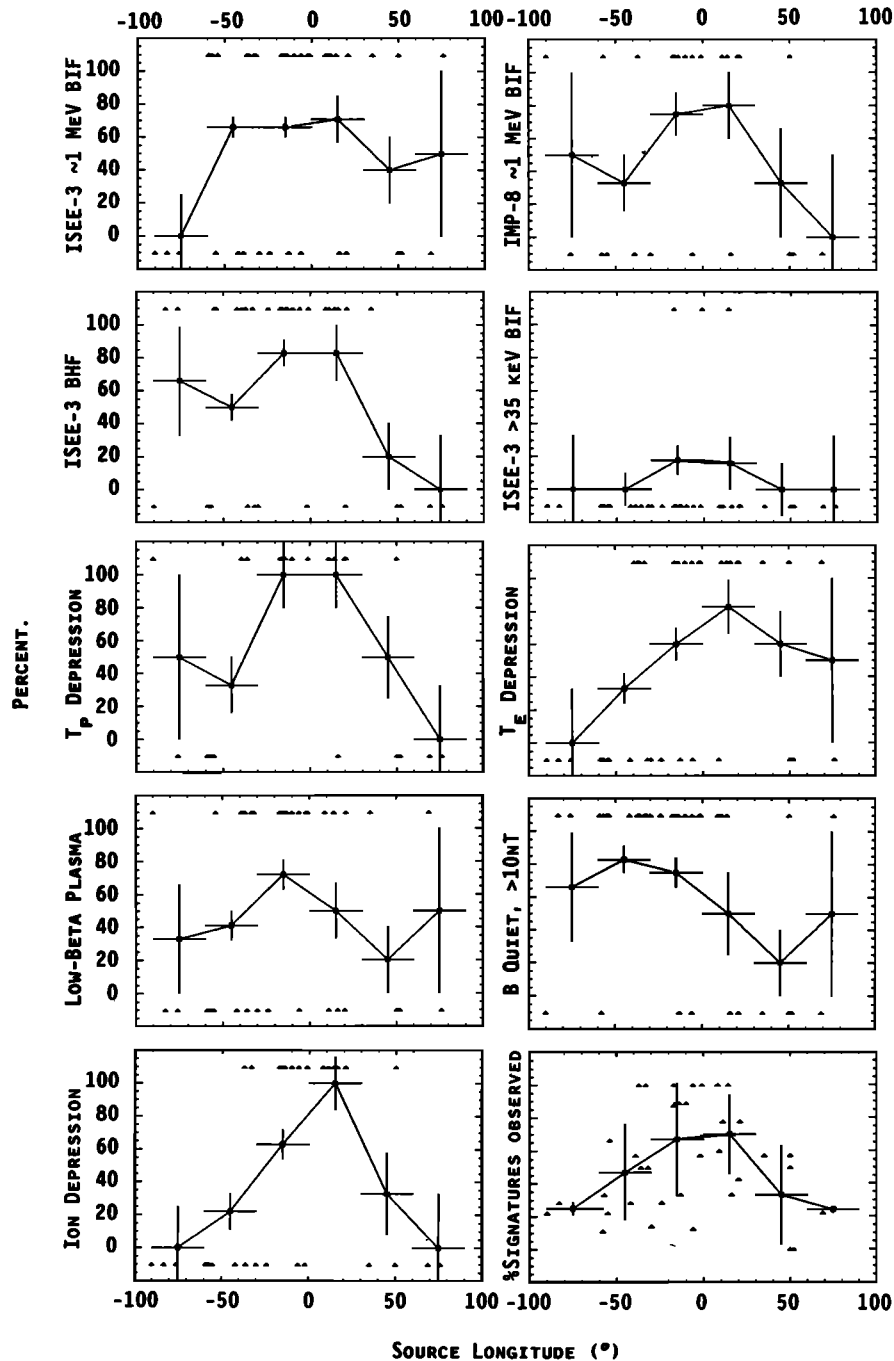


Fig. 1. Dependence of the occurrence of shock driver signatures on solar source longitude for 40 shocks with well-established source locations. Positive (negative) longitudes are west (east) of central meridian. A triangle at the top (bottom) of each panel indicates whether the signature was observed (not observed) commencing within ~24 hours following each shock. The mean occurrence rates (in percent) for each signature for events grouped in 30° bins in heliolongitude are also shown. (Errors indicate the effect of changing the number of events with a signature by one.) In general, these signatures are observed following shocks originating $<50^\circ$ from central meridian and are usually absent from other events, suggesting that shock drivers extend up to $\sim 50^\circ$ in longitude from the source. The bottom right panel shows the percentage of driver signatures (for which data are available) observed following each shock. The largest percentages of signatures are detected following central meridian events.

Considering shocks from sources around central meridian (which are expected to be followed by a driver), the percentages of these shocks associated with driver signatures are around 70% with bidirectional ~ 1 MeV/amu ions at ISEE 3, $\sim 80\%$ with bidirectional ~ 1 MeV/amu ions at IMP 8, 85% with solar wind electron bidirectional heat fluxes, $\sim 100\%$

with plasma proton temperature depressions, 80% with electron temperature depressions, $\sim 70\%$ with low- β plasma, $\sim 60\%$ with quiet, enhanced magnetic field (which is found more frequently following shocks with eastern sources), and $\sim 90\%$ with ion intensity depressions. However, we note that only $\sim 20\%$ of shocks from central meridian are followed by

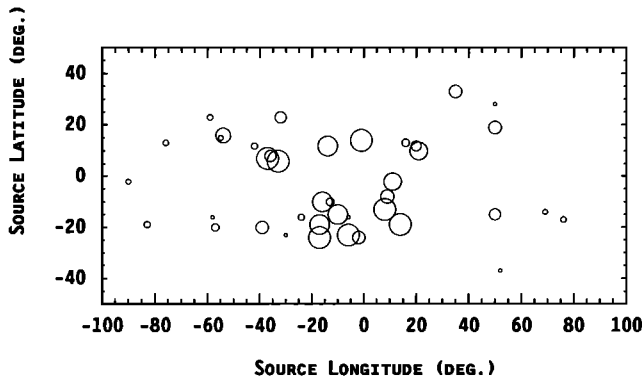


Fig. 2. Shock solar source locations plotted in terms of the longitude and latitude with respect to Earth. The size of the circle at each location is proportional to the number of shock driver signatures observed divided by the number of signatures for which data are available. Shocks from sources within $\sim 50^\circ$ of the Sun-Earth line are followed by most shock-driver signatures, and events from near-central meridian with relatively few driver signatures are not associated with sources with unusually high latitudes.

35–1000 keV bidirectional ion events reported by Marsden *et al.* [1987].

We also show in the bottom right-hand panel of Figure 1 the percentage of signatures for which data are available which are actually observed following each shock. Shocks from sources around central meridian show the largest number of signatures, with progressively fewer signatures being apparent with increasing source distance from central meridian. Around 70% of the signatures for which data are available are observed following shocks from near-central meridian events, falling to $\sim 20\%$ following shocks originating beyond 50° from central meridian. We note however that a few shocks from near-central meridian do not exhibit various shock driver signatures.

We have not considered the presence of enhanced solar wind helium abundances since published reports of enhanced helium abundances [Borini *et al.*, 1982; Zwickl *et al.*, 1983; Neugebauer and Alexander, 1991] cover only a small subset of the shocks considered here. Helium abundance enhancements have been reported following three of the shocks in Table 1, indicated by “He” in the last column. All of these shocks are from sources within 14° of central meridian and associated with a particularly high percentage of other driver signatures. Thus the association of shock drivers with He enhancements suggested by previous studies is supported with these observations.

The latitude of the solar source does not significantly influence our conclusions regarding the extent of shock drivers. Figure 2 shows the source locations plotted in terms of the longitude from central meridian and the latitude of the source relative to Earth (i.e., also taking the $\pm 7^\circ$ excursion of Earth relative to the solar equator into account). The size of the circle is proportional to the percentage of driver signatures (with data) observed. Events associated with large numbers of shock driver signatures are clearly produced by sources within $\sim 50^\circ$ of the Sun-Earth line, consistent with the conclusions drawn from considering only the longitudinal dependence. Figure 2 also indicates that the nondetection of shock driver signatures following a few

shocks from near-central meridian is not because these shocks originate from unusually high-latitude sources.

We also note in Table 1 that the February 4, 1983, shock was followed by an interval of clear bidirectional energetic ion streaming observed by ISEE 3 in the magnetosheath at $220 R_E$ downtail of Earth. This is believed to be the first observation of bidirectional ion flows in the deep-tail magnetosheath. Presumably, the bidirectional ion flows observed following this near-central meridian shock were associated with shock driver plasma, containing a population of bidirectional particles, which passed through the bow shock. Unfortunately, IMP 8 was not in the solar wind upstream of the bow shock and hence could not confirm that the bidirectional ion flows were indeed present before the shock driver encountered the bow shock and therefore were not confined to the deep-tail magnetosheath.

4. EXAMINATION OF INDIVIDUAL EVENTS

In this section we discuss three events from Table 1. Figure 3 shows solar wind plasma and energetic particle observations in the vicinity of a shock, initiated by a solar event near central meridian ($E14^\circ$), which produced an SC at 2302 UT on May 17, 1981, and was followed by most of the driver signatures included in Table 1. The top panel indicates intervals of field-aligned ISEE 3 and IMP 8 BIFs (identified using the method of Richardson and Reames [1993]) and solar wind electron BIFs (J. T. Gosling, private communication, 1989). No 35–1000 keV bidirectional ion interval was reported by Marsden *et al.* [1987] during the period shown in Figure 3. The next panel shows the 1–4 MeV ion intensity at ISEE 3, while the third to sixth panels display the magnetic field intensity and polar and azimuthal angles at ISEE 3, together with $\sigma B/B$. The bottom five panels show solar wind plasma parameters. The plasma proton temperature T_p , from the OMNI data base, is compared with the temperature expected from the well-established association between the solar wind speed and proton temperature [Lopez, 1987]. The plasma electron temperature T_e is from the ISEE 3 plasma experiment. The next panel shows the ratio β of the plasma particle (proton and electron) pressure to the magnetic field pressure. Where no OMNI proton temperature data are available, only the electron pressure is included in the calculation of β . Finally, the plasma density and solar wind speed from the OMNI data base are shown.

The shock is indicated by the solid vertical line in Figure 3. We identify the driver of this shock with the postshock plasma region extending from ~ 0700 UT to ~ 2200 UT on May 18 (indicated by vertical dashed lines). This is characterized by an enhanced (>10 nT), quiet ($\sigma B/B < 10^{-2}$) magnetic field. The proton temperature is clearly anomalously low in relation to that expected based on the solar wind speed and falls abruptly on entrance to the driver from the postshock region (where the temperature is higher than expected, presumably because of heating by the shock). The electron temperature is also depressed in the driver (to $\sim 10^5$ K), and β is low (<0.2). The 1–4 MeV ion intensity shows a depression in the same region, and field-aligned BIFs are observed at IMP 8 and ISEE 3 in the center of this region, close to the ion intensity minimum and also to the minimum in β . A bidirectional solar wind electron heat flux event commences close to the start of this region, also consistent with the shock driver association. However, it apparently

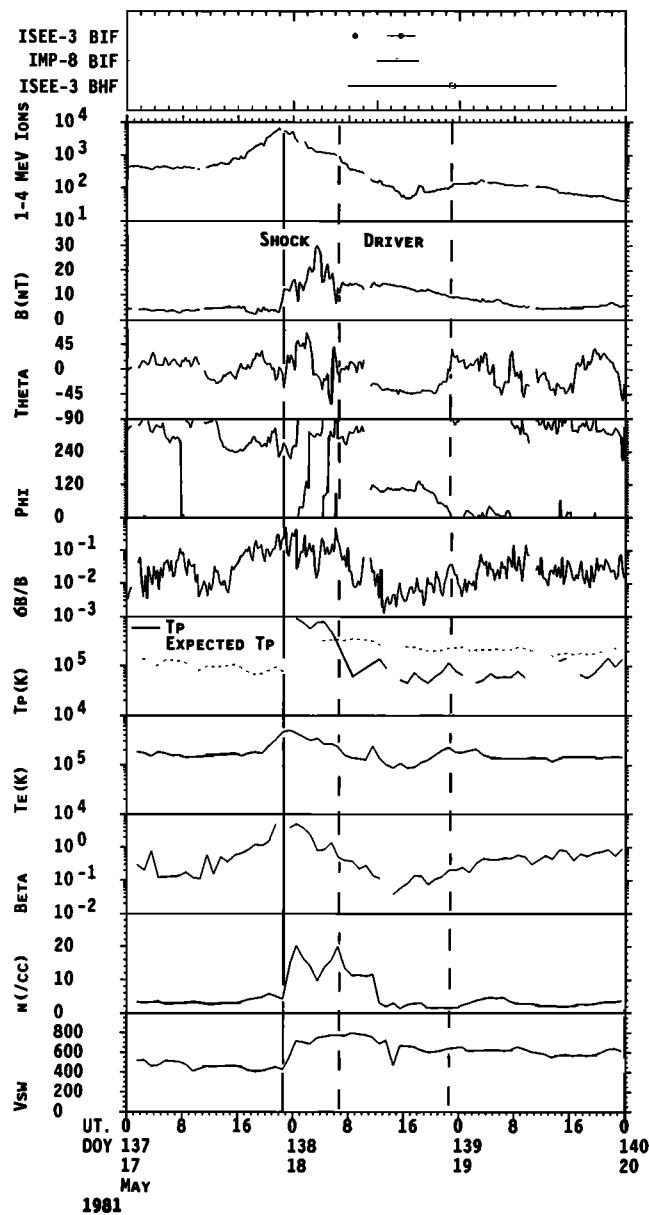


Fig. 3. Energetic ion, magnetic field, and solar wind plasma observations in the vicinity of the shock on May 17, 1981 (originating from a solar event at E16°), followed by several shock driver signatures, e.g., bidirectional ~ 1 MeV/amu ion flows (BIFs) at IMP 8 and ISEE 3; a solar wind electron bidirectional heat flux (BHF); depressed energetic ion intensity; quiet, enhanced magnetic field; plasma proton temperature depression below that expected from the solar wind speed; plasma electron temperature depression, and low- β plasma. The dashed vertical lines indicate the probable boundaries of the shock driver.

extends beyond the trailing edge of the inferred shock driver into a region of continuing cool plasma but with more variable, near-radial magnetic field. Thus it is possible that the driver may extend later than is indicated in Figure 3, although most other driver signatures (with the exception of the proton temperature depression) are not particularly evident.

Figure 4 shows a contrasting event: the shock at 0445 UT on March 4, 1979, associated with a solar event at E58°. Here there are no distinct driver signatures in the postshock

plasma. In particular, the proton temperature is close to (if anything, slightly exceeds) that expected from the solar wind speed, indicating that unusually cool plasma was not detected. Also, there is no evidence of a depression of T_e or of low- β plasma. Thus on the basis of the plasma data the driver was probably not encountered following this shock.

However, intervals of bidirectional energetic ion streaming were observed by ISEE 3 following the shock, as indicated in Figure 4. While bidirectional ion streaming following shocks is often associated with shock drivers, this is apparently not the case here: In addition to the absence of plasma shock driver signatures we note that the longest

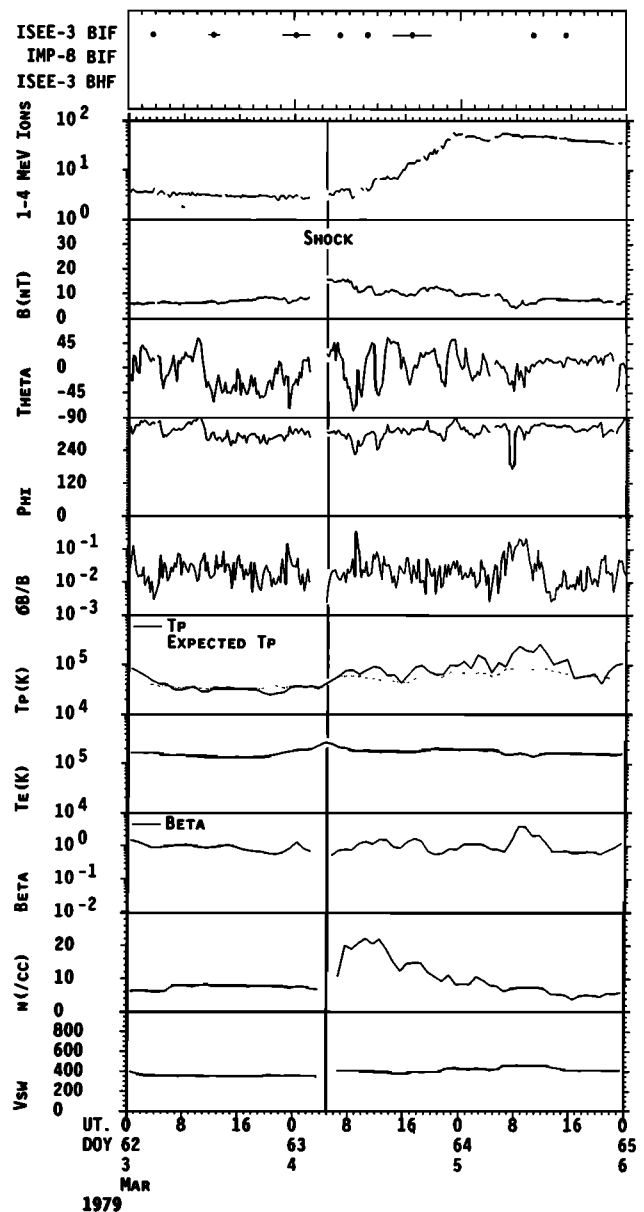


Fig. 4. Energetic ion, solar wind plasma, and magnetic field observations in the vicinity of the shock on March 4, 1979, from E58°. There are no driver signatures following this shock with the exception of bidirectional ~ 1 MeV ions at ISEE 3. The rise in ion intensity, often seen with shocks from far eastern events, suggests that the bidirectional ions are observed on postshock magnetic field lines linking to the nose of the shock rather than in the shock driver (which is not apparently encountered).

interval of bidirectional ion streaming occurs during an increase in the energetic ion intensity following the shock. An ion intensity increase often follows the passage of a shock from an eastern source [Cane *et al.*, 1988] of which this is an example. As shown in Figure 5 [after Cane, 1988; Gosling and McComas, 1987], field lines in the postshock region of a shock initiated by a source east of the spacecraft are connected to the strongest regions of the shock close to the nose. This can result in relatively high ion intensities in the postshock region. (In contrast, depressed ion intensities are usually observed inside shock drivers.) If this interpretation is correct, it implies that the postshock bidirectional ion flows in Figure 4 occurred on open field lines outside the shock driver.

Solar wind plasma and energetic ion observations for another shock from an eastern (E76°) event, the shock at 0953 UT on July 11, 1982, are shown in Figure 6. As in Figure 4, there are no shock driver signatures in the plasma data following the shock. In particular, the proton temperature is similar to, or exceeds, that expected from the solar wind speed, and there are no depressions in T_e or β . However, a bidirectional solar wind electron heat flux is observed. This occurs predominantly in a region of quiet and enhanced, but declining, magnetic field intensity following the shock, which resembles the magnetic field enhancement associated with the shock driver in Figure 3. Thus the bidirectional electron heat fluxes and the region of enhanced magnetic field together might suggest that the driver was encountered. However, the plasma data suggest otherwise. The postshock energetic ion intensity increase in this same region also suggests that the spacecraft was not in a shock driver but was on field lines connected to the nose of the shock. The ion anisotropies support this interpretation: Figure 7 shows 1–4 MeV ion anisotropies in the solar wind frame obtained during 2-hour intervals on July 11–12 (day of year (DOY) 192–193). Ion intensities in eight azimuthal

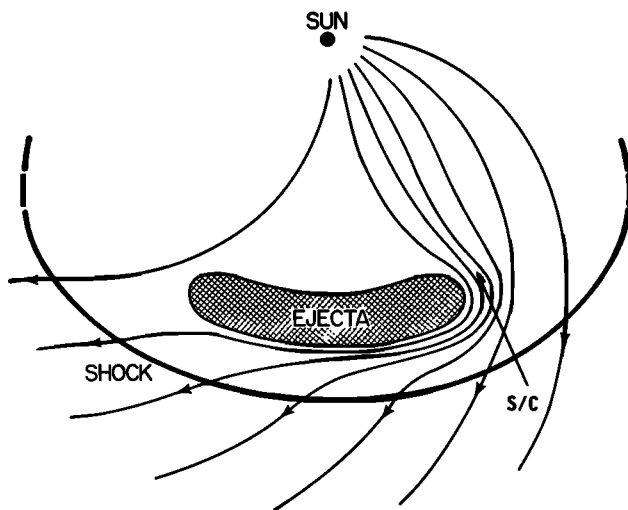


Fig. 5. Configuration of shock and shock driver (here represented as a "plasmoid," disconnected from the Sun) [after Cane, 1988; Gosling and McComas, 1987]. Magnetic field lines encountered by a spacecraft in the postshock region of a shock from an eastern event are connected to the strongest regions of the shock close to the nose of the shock. Because particle acceleration is most efficient at the nose of the shock, high energetic ion intensities can be observed by the spacecraft in the postshock region.

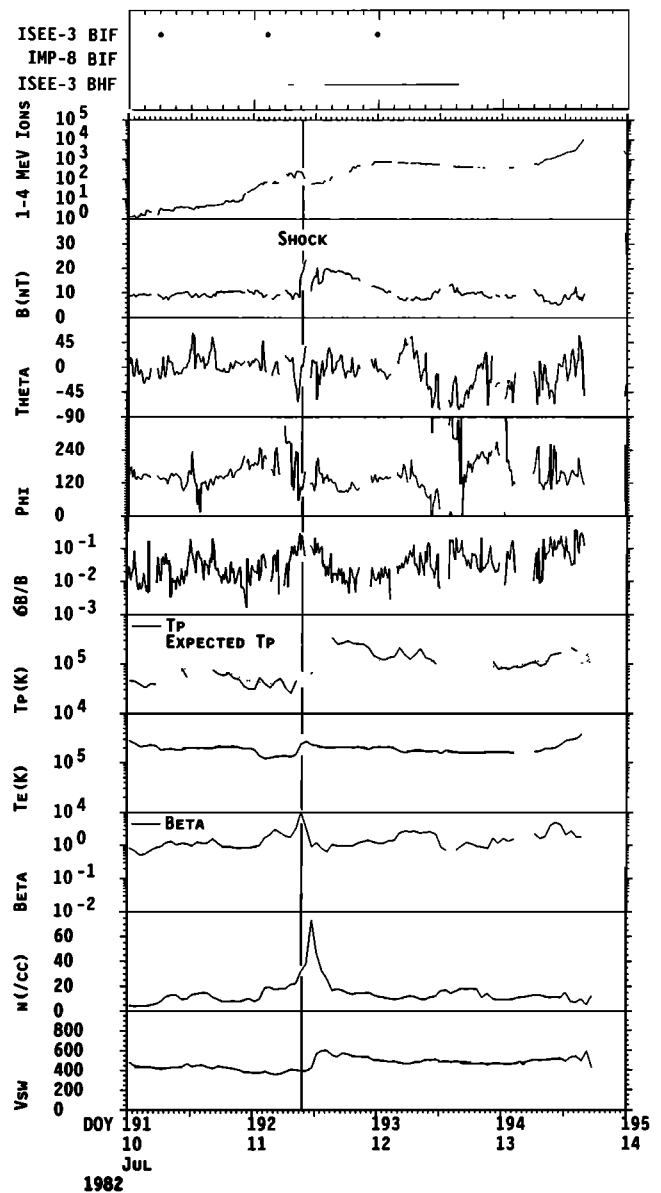


Fig. 6. Solar wind plasma, magnetic field, and energetic ion observations in the vicinity of the shock on July 11, 1982, from E73°. Driver signatures are generally absent. In particular, there are no plasma proton or electron temperature depressions or low- β plasma. However, a bidirectional solar wind electron heat flux and a region of enhanced, quiet magnetic field are observed following the shock. These features occur on postshock field lines linking to the nose of the shock, based on the enhancement of strongly sunward streaming ions observed during the heat flux event and magnetic field enhancement (see Figure 7).

sectors about the spacecraft are plotted versus viewing direction with the Sun to the top of Figure 7. The number of counts in the maximum count sector is indicated together with the variation in the direction of the magnetic field azimuth during each interval. Following the shock, the energetic ions are streaming sunward from ~1500 UT on July 11 (DOY 192). The sunward streaming is consistent with acceleration at, and flow from, the nose of the shock lying farther from the Sun and to the east of the spacecraft. The only evidence of bidirectional ion flows is during a brief interval at ~2330 UT on July 11, which is indicated in the top

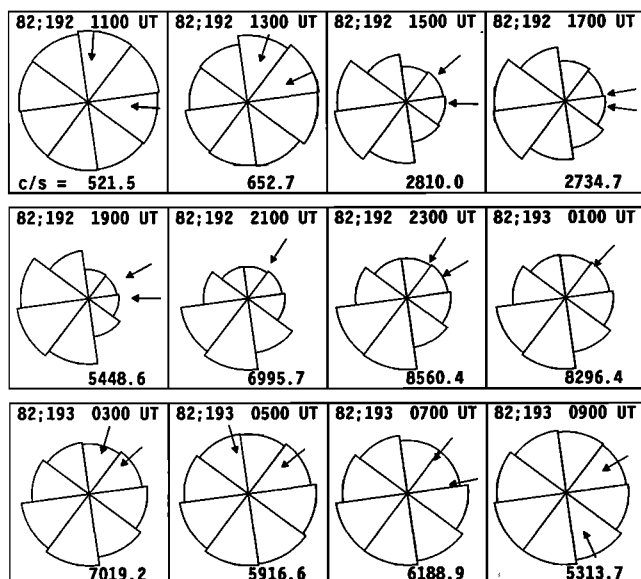


Fig. 7. Two-hour averaged 1–4 MeV ion anisotropies from the GSFC ISEE 3 instrument following the shock of July 11, 1982. Ion intensities in eight azimuthal sectors about the spacecraft, transformed into the solar wind frame, are plotted versus viewing direction with the Sun to the top. The number of particle counts per second in the maximum count sector and the variation in the magnetic field azimuth ($\pm 1\sigma$) are indicated. Sunward field-aligned streaming is observed in the postshock region from ~ 1500 UT on July 11 (DOY 192).

panel of Figure 6 but is not significant enough to be evident in the related 2-hour-averaged ion distribution.

5. SUMMARY AND DISCUSSION

We have examined the occurrence of shock driver signatures, in particular, magnetic field-aligned bidirectional energetic ion and solar wind electron heat fluxes, plasma proton and electron temperature depressions, low- β plasma, low-variance enhanced magnetic field, and energetic ion intensity depressions, following 40 energetic interplanetary shocks with well-established solar sources. Several driver signatures are commonly found following shocks from sources within $\sim 50^\circ$ of central meridian and are generally absent following other shocks. The observations suggest that shock drivers often extend up to $\sim 100^\circ$ in longitude centered on the solar source longitude.

A particular driver signature is typically observed following $\sim 70\%$ (but not all) of shocks from near-central meridian sources. However, only $\sim 20\%$ of central meridian shocks are followed by bidirectional 35–1000 keV ion events reported by Marsden *et al.* [1987] even though other shock driver signatures are usually observed following these shocks. In particular, $\sim 70\%$ of these shocks are associated with ~ 1 MeV/amu bidirectional ions from the GSFC instruments on ISEE 3 and IMP 8. It is unclear whether this discrepancy is due to a physical or instrumental reason (for example, Marsden *et al.* [1987] noted that bidirectional ions were more evident in their data at 600–1000 keV than at 35–56 keV) or whether data selection criteria are primarily responsible (for example, Marsden *et al.* [1987] reported bidirectional ion intervals with durations > 3 hours, while we have considered intervals of > 30 min duration). A detailed

investigation of this subject is beyond the scope of this study and would not affect our conclusions. The main point to note, however, is that though the reported 35–1000 keV bidirectional ion flows associated with shocks in this study show an excellent association with shock drivers, the absence of such flows following a shock according to the list of Marsden *et al.* [1987] should not be interpreted as proof that bidirectional ion streaming was not encountered.

We also note that although many of the shocks in the present study are followed by regions of quiet, enhanced magnetic field, none of the shocks are associated with the distinctive “magnetic clouds” (characterized by a magnetic field rotation with a magnetic field enhancement > 10 nT) identified by Klein and Burlaga [1982], Zhang and Burlaga [1988], and Lepping *et al.* [1990]. Hence none of the particularly energetic shocks (i.e., those associated with interplanetary type II radio emission) included in this study are driven by ejections which would be classified as magnetic clouds.

For a shock driver to be observed, it must extend in longitude over at least the angular separation between the solar source and Earth. An estimate of the distribution of shock driver longitudinal widths can therefore be made by considering how the probability of a shock being followed by a shock driver signature varies with source longitude. Figure 8 shows how the probability of detecting any of the shock driver signatures in Table 1 varies as a function of source longitude (solid circles), normalized to the probability at $\pm 10^\circ$ from central meridian. The rapid falls in the probability of detection at $\sim \pm 50^\circ$ from central meridian indicate that shock drivers typically extend to this distance eastward and westward from the source and that narrower or wider drivers are less frequent. We also show for comparison a distribution (open squares) based on the distribution of limb CMEs angular sizes (in latitude rather than longitude) observed by the SMM coronagraph [St. Cyr and Burkepile, 1990] but with the angular width doubled. This distribution shows reasonable agreement with the shock driver distribution. Assuming

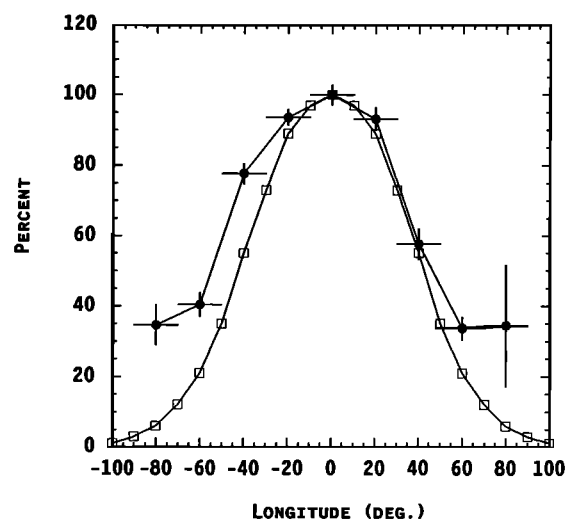


Fig. 8. Rate of detection of shock driver signatures versus source longitude (solid circles), normalized to the rate at 10° E to 10° W, providing an estimate of the probability of shock driver half widths exceeding the source longitude angle. This is compared with a distribution based on the occurrence rate of CME latitude extents from St. Cyr and Burkepile [1990] but with the angular span doubled (open squares).

that the longitudinal and latitudinal extents of CMEs are similar close to the Sun, it might be deduced from this comparison that CMEs expand during transit to 1 AU. However, *Cane et al.* [1987] found that the CMEs associated with very energetic shocks (in fact, a subset of the *Cane* [1985] shocks discussed here) have much larger angular extents than the majority of CMEs. This comparison suggests that CMEs do not expand between the Sun and 1 AU.

There are a few cases in Table 1 where driver signatures are observed following shocks from sources far from central meridian. One of these cases, the October 10, 1981, shock from E83°, was discussed by *Richardson et al.* [1991]. They concluded that the bidirectional solar wind electron heat flux (also associated with a region of quiet enhanced magnetic field) following the shock was not associated with this shock but was related to a steepening shock, probably from an unidentified central meridian source, which arrived at ~0700 UT on October 11. Hence the unexpected presence of shock driver signatures following shocks from near-limb events may result from the chance association of a weak shock from an event far from central meridian and a driver (or more generally, CME material) from another solar event originating nearer to central meridian.

In addition, we have discussed above two shocks from events well to the east of central meridian, where the plasma parameters show no evidence of a shock driver but bidirectional particle flows are observed. These bidirectional flows are apparently detected on field lines in the postshock region connected to the nose of the shock lying east of the spacecraft. There are several other events in Table 1 (including the October 4, 1978, shock from an E57° source, for which ion anisotropy observations are discussed by *Richardson and Reames* [1991]) which show bidirectional particle flows but little evidence of plasma shock driver signatures. These are predominantly eastern events, most followed by increases in ion intensity. (An exception is the shock of March 15, 1979, associated with a W76° source. However, the postshock plasma is complex, and it is possible that the (rather weak, intermittent) bidirectional ion flows observed by ISEE 3 are associated with structure unrelated to the shock.) They are also usually associated with quiet magnetic enhancements. Indeed, Figure 1 shows that there is a tendency for quiet field enhancements to follow shocks from eastern events, suggesting that they may not be uniquely associated with shock drivers.

Thus our observations suggest that in some eastern events the bidirectional particle flows and enhanced magnetic field following the shock may occur on open field lines in the postshock region, rather than in the shock driver. The field enhancement may be the result of the sweeping up of field lines around the western end of the driver (Figure 5), rather than be associated with the shock driver. (The absence of quiet magnetic field enhancements following shocks from far western events is also consistent with this proposal.) It is not immediately clear how bidirectional particle flows may develop in this geometry and why they may be present in some particle populations but not in others. However, we note that magnetic field configurations in which field lines converge may mirror energetic particles [*Barouch and Burlaga*, 1976]. Thus bidirectional flows may be set up by reflection between the Sun and the magnetic field compression associated with the shock, as discussed by *Rao et al.* [1967] and *Kahler and Reames* [1991]. The field enhancement associ-

ated with field line draping around the western edge of the shock driver may produce an additional magnetic mirror on these field lines. In this case the possibility exists of particle trapping and mirroring between this mirror and both the Sun and the shock front. The presence of bidirectional flows will then depend on the strength of this mirror on individual field lines and on the degree of particle scattering in each of these regions. We also note from *Behannon et al.* [1991] that the structure of the solar wind can be complex, involving many interacting transient and corotating flows. Thus magnetic mirrors may be created during the interaction of such flows, often remote from the spacecraft, resulting in trapped particle populations in addition to those clearly associated with shock drivers. It is unclear, however, whether bidirectional solar wind electron heat fluxes can be set up in a similar manner since they are believed to be produced when both ends of a field line are connected to a source of hot electrons, such as the Sun, or on closed field lines [*Gosling et al.*, 1992].

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